MOHAMED RUSAICK MUFTHI

Aspiring Cyber Security Analyst | Python, Networks & AI-driven Security

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PROFESSIONAL SUMMARY

HND Software Engineering graduate currently reading for a BSc (Hons) Computer Networks top-up degree (University of Wolverhampton, via CINEC Campus), built deliberately to transition from software development into Cyber Security. Final-year research focused on ML-based detection of XSS attacks, reaching 99.81% classification accuracy across 5 models. Comfortable across Python, networking fundamentals, and applied AI, with a track record of independently designing, building, and evaluating full security-relevant systems end to end. Former President of the BCAS IT Society, experienced in leading teams and driving technical initiatives.

CORE SKILLS

- **Security & Networks:** Web application security (OWASP, XSS), threat/attack classification, security fundamentals, networking concepts (in progress via BSc Computer Networks)
- **Programming:** Python (primary), SQL/SQLite, HTML, CSS
- **AI/ML for Security:** Scikit-learn, XGBoost, TensorFlow/Keras, Logistic Regression, Random Forest, feature engineering, model evaluation (Accuracy, Precision, Recall, F1, ROC-AUC)
- **Tools & Platforms:** Streamlit, Jupyter Notebook, REST APIs, Pandas, NumPy, Matplotlib
- **Other:** Clean/documented code, cross-functional team leadership, English (professional), Tamil (native)

PROJECTS & RESEARCH

XSS Attack Detection — ML vs Neural Networks Jan 2025 – Sep 2025

HND Final-Year Research Project

- Independently researched and benchmarked 4 classical ML classifiers (Logistic Regression, Naïve Bayes, Random Forest, XGBoost) against a Multi-Layer Perceptron for detecting Cross-Site Scripting (XSS) attacks, a critical OWASP web security threat.
- Preprocessed a labelled HTTP payload dataset, engineered features, and evaluated all 5 models across Accuracy, Precision, Recall, F1-score, and ROC-AUC, writing all evaluation code from scratch.
- XGBoost achieved 99.81% accuracy and 0.999 ROC-AUC, demonstrating that well-tuned tree-based models can match neural networks for structured security classification at lower compute cost.
- Built a real-time Streamlit dashboard for live payload input, instant classification, and interactive model comparison — directly applicable to SOC-style triage tooling.
- **Stack:** Python · Scikit-learn · TensorFlow/Keras · Pandas · Matplotlib · Streamlit

AI-Powered Hospital Queue Management System Sep 2024 – Dec 2024

Academic Project

- Built an end-to-end data pipeline and web application, handling data ingestion, feature engineering, model training, and real-time prediction in a single deployable system — the same engineering discipline applied to building reliable security tooling.
- Trained and benchmarked 3 regression models, selecting Decision Tree for the best R^2 score (0.915) on held-out test data.
- Integrated SQLite for persistent records and the Twilio API for automated notifications, demonstrating secure handling of data pipelines and third-party API integration.
- **Stack:** Python · Scikit-learn · XGBoost · Streamlit · SQLite · Pandas · NumPy · Matplotlib · Twilio API

EDUCATION

BSc (Hons) Computer Networks (Top-Up) Oct 2026 - Present

CINEC Campus, Sri Lanka — affiliated with University of Wolverhampton, UK

- Pursuing specialized study in networking, systems administration, and security principles to build toward a Cyber Security career.

Pearson BTEC HND in Software Engineering Jan 2024 – Sep 2025

BCAS Campus, Kalmunai, Sri Lanka

- Relevant modules: Machine Learning · Data Analytics · Software Engineering Principles · Database Systems · UI/UX Design

Ordinary & Advanced Level 2019 – 2023

Zahira College, Kalmunai, Sri Lanka

CERTIFICATIONS

- LinkedIn Learning: AI for Cybersecurity · Career Skills in Software Development
- Simplilearn: Cyber Security · Machine Learning using Python · Cloud Computing
- AWS: Educate Machine Learning Foundations
- Udacity: Introducing Generative AI with AWS
- Sololearn: Python Developer · Generative AI in Practice · Ethical AI Foundations · Web Development
- Great Learning: Python for ML & Data Science · Java Programming

LEADERSHIP & VOLUNTEERING

President — BCAS IT Society (BITS) Apr 2025 – Nov 2025

BCAS Campus Kalmunai, Sri Lanka

- Led a 60+ member society; organized 4+ tech workshops reaching 200+ students; drove team coordination and student engagement.

Treasurer — BCAS IT Society (BITS) May 2024 – Apr 2025

BCAS Campus Kalmunai, Sri Lanka

- Managed financial planning and budgeting for all society events; maintained accurate, transparent financial records.

REFERENCES

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